

Modules 4b – 7a Sample Questions

Allowed time: 90 minutes

What is this collection of sample questions?

This exam contains a collection of questions used on previous Chemistry 103 exams from a variety of instructors. Each learning objective is addressed by at least one question, and the learning objectives are listed on the key.

This collection of sample questions matches the structure of the exam since it contains each type of question format: multiple choice, fill in the blank, and short answer questions. This collection also reflects what you have worked on in pre-class activities, whole class activities, OWL homework, and in the laboratory.

This collection of sample questions does not match the structure of the exam since it will take longer than your allotted class time. Thus, the sample exam provides *more* questions than you'll see on an actual exam. Your actual exam will not address every learning objective, and your actual exam will be allotted 75 minutes.

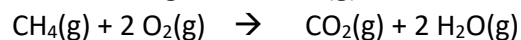
How to use this exam as a study tool:

Go to a quiet place and take this exam under *real* exam conditions: Work alone with only your pencil, calculator, notecard, attached reference sheet, and attached periodic table as aids.

After completing the exam, check your work compared to the answer key. If you get a question incorrect, then seek help from your whole class notes, textbook, pre-class activities, lab manual, or friends. Other good resources include your TA/FA, instructor, and the Chem 103 Help desk.

Chemistry 103 Exam 2 Sample Exam Key

1. The combustion of methane releases 891 kJ of heat. How much heat is released by the combustion of 2.00 grams of $\text{CH}_4(\text{g})$?

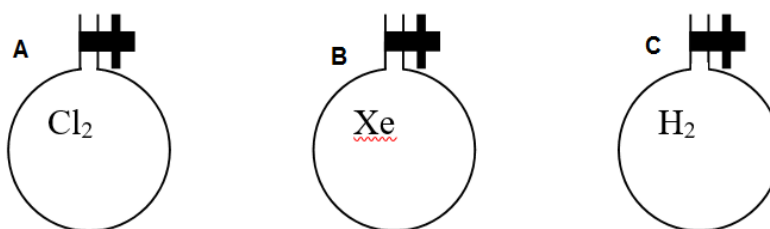


- a) 891 kJ
 - b) 1780 kJ
 - c) 111 kJ
 - d) 446 kJ
2. Which of the following processes are exothermic:
- 1) Adding magnesium to a solution of HCl
 - 2) Water freezing on a cherry blossom
 - 3) The reaction of H_2 and O_2 gases to form water
- a) all of the above
 - b) 2 and 3 only
 - c) 1 and 2 only
 - d) 1 and 3 only
 - e) none of the above
3. To prepare a 0.10 M solution of K_2CrO_4 , you add 1000. mL of water to 100. mL of a 1.00 M solution of K_2CrO_4 . True or false?
- a) true
 - b) false
4. A 25.00 mL sample of KOH is titrated with 28.68 mL of 0.2120 M $\text{HNO}_3(\text{aq})$ to reach the equivalence point. What is the concentration of the KOH solution?
- a) 0.0006720 M
 - b) 0.002574 M
 - c) 0.1851 M
 - d) 0.2432 M
 - e) 4.119 M

5. In the “collapsing can” experiment demonstrated in class (where a small amount of water was added to an empty soda can and brought to a boil for several minutes) which of the following statements is **false**?
- Before heating, the pressure of gas inside the open can was the same as the pressure outside of the open can.
 - When the water was boiling in the can, the pressure of gas inside the open can was the same as the pressure outside of the open can.
 - When placed in the cold water, the water vapor in the can condensed to liquid water.
 - The vacuum created in the can pulled the walls of the can inwards.
 - Atmospheric pressure pushed the walls of the can inwards.
6. Which of the following variables for a gas is/are **inversely related** to each other?
- Pressure and Volume
 - Pressure and Temperature
 - Pressure and Number of Moles
- a) (i) Only b) (ii) Only c) (iii) Only d) (i) and (ii) Only e) (ii) and (iii) Only

Answer the following two questions regarding the gases in the flasks below. You can assume each gas behaves as an ideal gas.

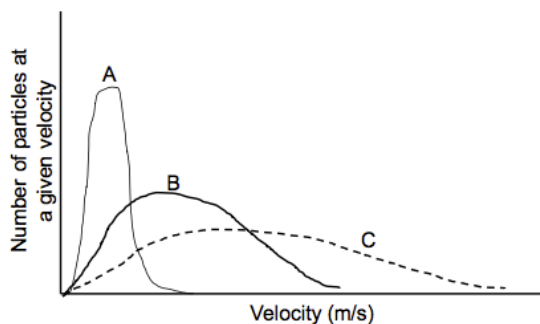
All flasks have the same conditions: volume is 2.50 L, pressure in each flask is 662 mmHg, and the temperature is 105.0 °C



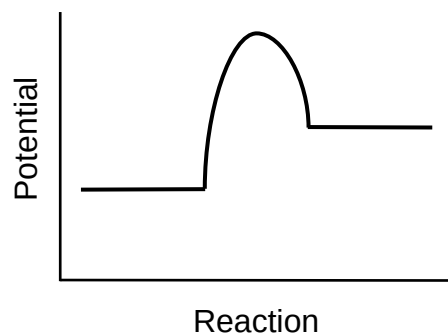
7. Which of the following statements are **true**:
- Each flask contains the same number of **moles** of gas.
 - Each flask contains the same **mass** of gas.
 - The rms speed, $\sqrt{v^2}$, for each gas is the **same**.
- a) (i) Only b) (ii) Only c) (iii) Only d) (i) and (ii) Only e) (ii) and (iii) Only
8. Determine the density of the Xe gas in Flask B:
- a) 7.41 g/L b) 0.0364 g/L c) 13.3 g/L d) 2.80×10^3 g/L e) 3.69 g/L

9. Which of the following is true for an ideal gas?
- a) The gas particles have a measurable volume.
 - b) All particles in a sample at 300 Kelvin are moving at identical speeds.
 - c) The attraction between gas particles is negligible.
 - d) The sample will be colorless

The following two questions refer to the graph below.



10. Fluorine, bromine, and chlorine gas are each contained in a separate, sealed vessel. All three vessels are at 298 K. Which line represents the speed distribution of **chlorine** gas particles?
- a) A
 - b) B
 - c) C
11. Three nitrogen gas samples are at three different temperatures (273 K, 1273 K, and 2273 K). Which line represents the speed distribution of nitrogen gas particles at **273 K**?
- a) A
 - b) B
 - c) C
12. Which of the following is a characteristic of a greenhouse gas?
- a) Absorbs infrared (IR) light.
 - b) Has a high density.
 - c) Absorbs green light.
 - d) Reacts readily with other gases.



13. The reaction represented by the graph above is:

- a) Endothermic
- b) Exothermic
- c) Cannot be determined from this graph

14. Two reagents react in a sealed, *insulated* calorimeter such that no heat can flow into or out of the calorimeter. During the process of reaction, the water inside the calorimeter gets colder.

- a) Consider the *system* to be the contents of the calorimeters (reagents and water), this process is:

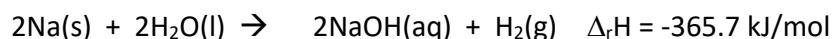
exothermic endothermic $q=0$ **(circle one)**

- b) Considering the *system* to be the reagents only (and *not* the water), and the *surroundings* to be everything else (*including* the water), this process is:

exothermic endothermic $q=0$ **(circle one)**

15. CaCl_2 dissolves in water exothermically. Which of the following experiments produces the largest increase **of thermal energy**?
- a) Dissolving 1.0 g CaCl_2 in 100 mL of water.
 - b) Dissolving 2.0 g CaCl_2 in 100 mL of water.
 - c) Dissolving 1.5 g CaCl_2 in 50 mL of water.
 - d) Dissolving 0.50 g CaCl_2 in 150 mL of water.
 - e) They would all produce the same quantity of thermal energy.
16. CaCl_2 dissolves in water exothermically. If the reaction takes place in a Styrofoam cup which of the following experiments would produce the largest **temperature change**?
- a) Dissolving 1.0 g CaCl_2 in 100 mL of water.
 - b) Dissolving 2.0 g CaCl_2 in 100 mL of water.
 - c) Dissolving 1.5 g CaCl_2 in 50 mL of water.
 - d) Dissolving 0.50 g CaCl_2 in 150 mL of water.
 - e) The temperature change would be the same in all cases.

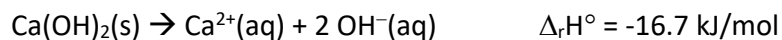
17. For the following chemical reaction that takes place at constant pressure:



Which of the following statements are **true**?

- (i) $q_p = -365.7 \text{ kJ}$
 - (ii) Work is done by the system on the surroundings.
 - (iii) $\Delta E = -365.7 \text{ kJ}$
- a) (i) Only
 - b) (ii) Only
 - c) (iii) Only
 - d) (i) and (ii) Only
 - e) (ii) and (iii) Only
18. Which of the following is not a state function?
- a) Temperature
 - b) Enthalpy
 - c) Work
 - d) Internal energy

19. From the following enthalpies of reaction



one calculates the $\Delta_r H^\circ$ for the reaction $\text{Ca}^{2+}(\text{aq}) + 2 \text{OH}^{-}(\text{aq}) + \text{CO}_2(\text{g}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$ to be:

- a) 260.2 kJ/mol
- b) 129.8 kJ/mol
- c) -260.2 kJ/mol
- d) -129.8 kJ/mol
- e) -96.4 kJ/mol

20. Select the **FALSE** statements related to the electromagnetic spectrum.

- i. Microwaves have longer wavelengths than x-rays
- ii. Radio waves have higher frequency than ultraviolet radiation
- iii. Red light has higher energy than blue light

- a) i only
- b) ii only
- c) iii only
- d) all statements are true
- e) i and ii
- f) i and iii
- g) ii and iii
- h) i, ii, and iii

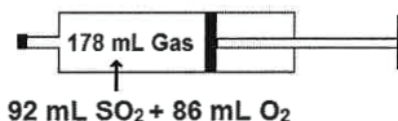
21. According to experiments concerned with the photoelectric effect, which of the following would increase the kinetic energy of an electron ejected from a metal surface?

- i. Increasing the number of photons of light striking the surface.
- ii. Decreasing the frequency of the light striking the surface.
- iii. Decreasing the wavelength of the light striking the surface.

- a) i only
- b) ii only
- c) iii only
- d) none of these
- e) i and ii
- f) i and iii
- g) ii and iii
- h) i, ii, and iii

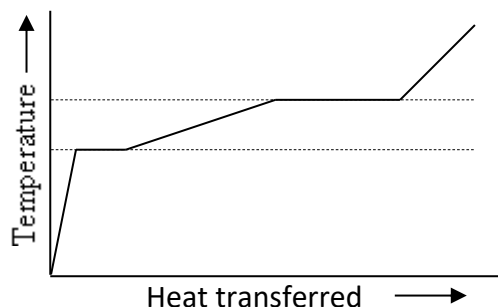
Short Answer and fill in the blank:

22. 92 mL of sulfur dioxide gas is mixed in a sealed syringe (with a movable plunger) with 86 mL of oxygen gas to give a total of 178 mL of gas.



- a) What volume does the O₂ gas occupy in the syringe once the two gases are mixed?
- b) Identify the gas or gases in the syringe after the sulfur dioxide and oxygen react to give the maximum amount of sulfur trioxide gas as the only product.
- c) Determine the final volume of gas in the syringe after the reaction. (Assume pressure and temperature remain constant).
- d) If immediately after the completion of the reaction of SO₂ and O₂ the volume of the gas in the syringe was **lower** than that calculated in (c), would this imply the reaction was exothermic or endothermic? Explain.

23. The following graph is the heating curve for compound X:



- a) On the **graph above**, CAREFULLY **circle** the region corresponding to vaporization of X. Ambiguous answers will not receive credit.

For the remaining questions, **circle one** of the choices below.

- b) Which has the lowest heat capacity: X(s) X(l) X(g)

- c) Which is smallest enthalpy change: $\Delta_{\text{fus}}H^\circ(\text{X})$ $\Delta_{\text{vap}}H^\circ(\text{X})$ $\Delta_{\text{sublimation}}H^\circ(\text{X})$

24. If the same amount of energy in the form of heat is added to 5.00 g samples of each metal listed below, which metal will undergo the largest temperature change?

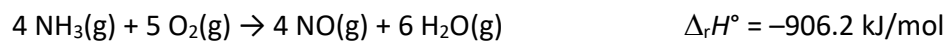
Metal	Specific Heat Capacity ($\frac{J}{gK}$)
Ni	0.440
Al	0.897
Cu	0.385
Fe	0.449
Mg	1.017

Metal: _____

25. Consider the reaction for the formation of nitric oxide, NO(g).

a) Write the complete, balanced equation for the formation of NO(g) from its constituent elements.

b) Use the information provided below to calculate the standard enthalpy of formation of NO(g), $\Delta_f H^\circ[\text{NO(g)}]$.



$\Delta_f H^\circ[\text{NO(g)}] = \underline{\hspace{2cm}}$